

NETWORK ANOMALY DETECTION SYSTEM

Phase 3 Implementation Report: VLAN Configuration & Inter-VLAN Routing

Project:	Network Anomaly Detection (ML + Network Engineering)
Team Lead:	Elina (P1) - Network Engineer & Team Lead
Report Date:	June 05, 2026
Status:	In Progress - Phase 3 (Debugging)
Environment:	Cisco Packet Tracer 8.2 Windows 11 PowerShell

EXECUTIVE SUMMARY

This report documents the third phase of the Network Anomaly Detection System project: manual configuration of VLANs, DHCP, and inter-VLAN routing in Cisco Packet Tracer. The project is a 4-person academic team effort combining network engineering, machine learning, backend development, and frontend dashboard work. Phases 1-2 (IP addressing and OSPF routing) were completed successfully. Phase 3 is currently under debugging, with DHCP operational but inter-VLAN traffic failing between HR-PC1 (VLAN 20) and DNS-WebServer (VLAN 99).

PROJECT OVERVIEW

Objective: Build a complete network anomaly detection system that combines network traffic capture, machine learning model training, Flask REST API backend, and interactive dashboard. The system processes the CICIDS dataset (labeled network attacks) to identify anomalies in real-time.

Team Structure:

- P1 (You): Network Engineer + Team Lead — Responsible for network topology, OSPF routing, VLAN configuration, and DevNet automation
- P2: Security/Data Engineer — Handles dataset security and CICIDS ingestion
- P3: ML Engineer — Builds anomaly detection model on processed data
- P4: Frontend/Dashboard Engineer — Creates web UI for anomaly visualization

Current Phase: Phase 3 — VLAN Configuration & Inter-VLAN Routing (Debugging)

Project Repository: github.com/elinayasmin/network-anomaly-detection

Interview Preparation: All work designed to demonstrate Cisco networking knowledge for interview with Cisco

COMPLETED MILESTONES (W1-W3)

Week 1-2: Flask API & Backend Setup ■

- Created REST API with 3 endpoints: GET /api/health, GET /api/anomalies, POST /api/anomalies
- Implemented SQLite database (api/anomalies.db) with schema: id, timestamp, src_ip, dst_ip, protocol, anomaly_score, label
- GitHub repository initialized with folder structure: capture/, ml/, api/, dashboard/, docs/

Week 3: CICIDS Data Ingestion ■

- Downloaded CICIDS dataset (Monday-normal, Tuesday-FTP/SSH attacks, Friday-PortScan)
- Created ingest.py pipeline: CSV → Extract attack context → Send to Flask API → Store in DB
- Successfully ingested 200+ attack records with labels (BENIGN, FTP-Patator, PortScan)
- Data validation: All 201 API responses successful, data persisted in SQLite

Phase 1-2: Packet Tracer Topology ■

- Designed 4-router backbone with point-to-point /30 links
- Configured IP addresses on all router interfaces and serial links
- Implemented OSPF routing (Area 0) with all routers peering in FULL state
- Verified routing with show ip route — all networks learned dynamically via OSPF
- Created 4 switches (SW-Floor1, SW-Floor2, SW-Floor3, SW-Servers) with VLAN definitions

NETWORK CONFIGURATION & IP ADDRESSING

1. Router Backbone (P2P Links)

Link	Subnet	R1 IP	R2 IP	R3 IP	R4 IP
R1-R2	192.168.0.0/30	192.168.0.1	192.168.0.2	-	-
R2-R3	192.168.0.4/30	-	192.168.0.5	192.168.0.6	-
R2-R4	192.168.0.8/30	-	192.168.0.9	-	192.168.0.10

2. VLAN Networks (User Subnets)

VLAN	Name	Network	Gateway IP	Router	Devices	Status
99	Management	192.168.1.0/28	192.168.1.1	R2 Gig0/0.99	DNS-WebServer	■
10	Floor 1 IT	192.168.2.0/26	192.168.2.1	R3 Gig0/0.10	IT-PC1, IT-PC2	■
20	Floor 2 HR	192.168.3.0/27	192.168.3.1	R4 Gig0/0.20	HR-PC1, HR-PC2	■■
30	Floor 3 Finance	192.168.4.0/27	192.168.4.1	R4 Gig0/1.30	FIN-PC1, FIN-PC2	■

Note: ■■ = Currently failing inter-VLAN routing (HR-PC1 cannot reach DNS-WebServer). See 'Issues Encountered' section.

VLAN CONFIGURATION DETAILS

Switch Configuration Summary:

Each switch (SW-Floor1, SW-Floor2, SW-Floor3, SW-Servers) is configured with:

- Access ports (Fa0/1-3): Assigned to respective VLAN (10, 20, 30, or 99)
- Trunk port (Fa0/4): Connects to router, native VLAN 99
- Trunk encapsulation: 802.1q (automatic on 2960 switches)
- Allowed VLANs: 1 (native), VLAN-specific, and 99 (management)

Router Subinterface Configuration:

Used router-on-a-stick model where one physical trunk port carries multiple VLANs. Each VLAN has a subinterface with encapsulation dot1Q:

R2 Configuration:

- interface Gig0/0 (trunk, no IP)
- interface Gig0/0.99: 192.168.1.1/28 (VLAN 99)

R3 Configuration:

- interface Gig0/0 (trunk, no IP)
- interface Gig0/0.10: 192.168.2.1/26 (VLAN 10)

R4 Configuration:

- interface Gig0/0 (trunk, no IP)
- interface Gig0/0.20: 192.168.3.1/27 (VLAN 20)
- interface Gig0/1 (trunk, no IP)
- interface Gig0/1.30: 192.168.4.1/27 (VLAN 30)

DHCP Configuration:

Each router running DHCP server for its VLAN(s):

- R2: DHCP pool "Management" → 192.168.1.0/28 with gateway 192.168.1.1
- R3: DHCP pool "Floor1-IT" → 192.168.2.0/26 with gateway 192.168.2.1
- R4: DHCP pools "Floor2-HR" and "Floor3-Finance" → 192.168.3.0/27, 192.168.4.0/27
- Excluded addresses: .1 - .5 (reserved for gateway and servers)
- DNS server: 8.8.8.8 (Google)

ISSUES ENCOUNTERED & DEBUGGING

Primary Issue: Inter-VLAN Routing Failure

HR-PC1 (in VLAN 20, IP 192.168.3.x) cannot ping DNS-WebServer (in VLAN 99, IP 192.168.1.10).

Result: 100% packet loss on ping attempts

What IS Working:

- DHCP: All user PCs obtained IPs in correct ranges (192.168.2.x, 192.168.3.x, 192.168.4.x)
- Same-VLAN communication: PCs in same VLAN can reach each other
- OSPF routing: All routers peering (FULL state), routes dynamically learned
- Router subinterfaces: All Gig0/0.xx interfaces UP with correct IPs
- Trunk port status: SW-Floor2 Fa0/4 and SW-Floor3 Fa0/4 showing "trunking"
- VLAN membership: Ports correctly assigned to VLANs

What is NOT Working:

- Cross-VLAN traffic: HR-PC1 cannot reach any other VLAN (100% loss)
- Router forwarding: No error messages, traffic silently drops
- No visible failure point: show ip route shows correct routes, interfaces UP

Suspected Root Causes (Not Yet Verified):

1. Trunk ports not properly forwarding VLAN-tagged frames to routers
2. HR-PC1 not configured with correct gateway (192.168.3.1)
3. R4 Gig0/0.20 not receiving packets from trunk
4. Asymmetric routing: forward path works but return path blocked
5. Switch spanning tree blocking trunk ports

Debugging Steps Taken:

- Verified all router subinterface IPs and status
- Verified VLAN assignments on all switches
- Verified OSPF routes and DHCP pools
- Reconfigured trunk ports with proper encapsulation and allowed VLANs
- Removed conflicting subinterfaces (Gig0/0.1, Gig0/0.30)
- Checked physical cable connections in topology

What Still Needs Debugging:

- Packet-level capture on router (debug ip packet detail)
- Traceroute from HR-PC1 to see failure hop
- Port counters on switch trunk to verify traffic arrival
- Verify HR-PC1 ipconfig shows gateway 192.168.3.1
- Check for spanning-tree port blocking on trunks

CONFIGURATION VERIFICATION SCREENSHOTS

Screenshot 1: R2 Interface Configuration & OSPF Routes

```
R2#show ip interface brief
Interface                               IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0                     192.168.1.1     YES manual up          up
GigabitEthernet0/0.99                   192.168.1.1     YES manual up          up
GigabitEthernet0/1                     unassigned      YES unset   administratively down down
GigabitEthernet0/2                     unassigned      YES unset   administratively down down
Serial0/2/0                             192.168.0.2     YES manual up          up
Serial0/2/1                             192.168.0.5     YES manual up          up
Serial0/3/0                             192.168.0.9     YES manual up          up
Serial0/3/1                             unassigned      YES manual down        down
Vlan1                                   unassigned      YES unset   administratively down down

R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

192.168.0.0/24 is variably subnetted, 6 subnets, 2 masks
C      192.168.0.0/30 is directly connected, Serial0/2/0
L      192.168.0.2/32 is directly connected, Serial0/2/0
C      192.168.0.4/30 is directly connected, Serial0/2/1
L      192.168.0.5/32 is directly connected, Serial0/2/1
C      192.168.0.8/30 is directly connected, Serial0/3/0
L      192.168.0.9/32 is directly connected, Serial0/3/0
L      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/28 is directly connected, GigabitEthernet0/0.99
L      192.168.1.1/32 is directly connected, GigabitEthernet0/0.99
C      192.168.2.0/26 is subnetted, 1 subnets
O      192.168.2.0/26 [110/65] via 192.168.0.6, 00:03:26, Serial0/2/1
```

Screenshot 2: R4 Subinterface Configuration & Routes

```
R2#show interfaces Gig0/0
GigabitEthernet0/0 is up, line protocol is up (connected)
Hardware is CN Gigabit Ethernet, address is 0002.1617.ab01 (bia 0002.1617.ab01)
Description: Ser- LAN
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s, media type is RJ45
output flow-control is unsupported, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00,
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue : 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
0 packets input, 0 bytes, 0 no buffer
Received 0 broadcasts, 0 runs, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 watchdog, 1017 multicast, 0 pause input
0 input packets with dribble condition detected
42 packets output, 2688 bytes, 0 underruns
0 output errors, 0 collisions, 1 interface resets
0 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out

R2#
```

Screenshot 3: R4 Gig0/0 & Gig0/1 Interface Details

```
R4#show ip interface brief
Interface                               IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0                     unassigned      YES manual up          up
GigabitEthernet0/0.1                   unassigned      YES unset up          up
GigabitEthernet0/0.20                  192.168.3.1     YES manual up          up
GigabitEthernet0/1                     192.168.4.1     YES manual up          up
GigabitEthernet0/2                     unassigned      YES unset   administratively down down
Serial0/3/0                             192.168.0.10    YES manual up          up
Serial0/3/1                             unassigned      YES unset   administratively down down
Vlan1                                   unassigned      YES unset   administratively down down

R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

192.168.0.0/24 is variably subnetted, 4 subnets, 2 masks
O00    192.168.0.0/30 [110/128] via 192.168.0.9, 00:12:58, Serial0/3/0
T000    192.168.0.4/30 [110/128] via 192.168.0.9, 00:12:58, Serial0/3/0
C000    192.168.0.8/30 is directly connected, Serial0/3/0
L000    192.168.0.10/32 is directly connected, Serial0/3/0
O      192.168.1.0/28 is subnetted, 1 subnets
C      192.168.1.0/28 [110/65] via 192.168.0.9, 00:05:44, Serial0/3/0
O      192.168.2.0/26 is subnetted, 1 subnets
C      192.168.2.0/26 [110/129] via 192.168.0.9, 00:04:34, Serial0/3/0
O      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
L      192.168.3.0/27 is directly connected, GigabitEthernet0/0.20
L      192.168.3.1/32 is directly connected, GigabitEthernet0/0.20
```

Screenshot 4: SW-Floor2 VLAN & Trunk Configuration

```

SW-Floor2>en
SW-Floor2#show vlan brief

VLAN Name                Status    Ports
----
1    default              active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
20   HR-Dept              active    Fa0/1, Fa0/2, Fa0/3
99   Management           active
1002 fddi-default          active
1003 token-ring-default  active
1004 fddinet-default     active
1005 trnet-default      active

SW-Floor2#show interfaces trunk

Port      Mode      Encapsulation  Status      Native vlan
Fa0/4     on        802.1q         trunking    99

Port      Vlans allowed on trunk
Fa0/4     1-1005

Port      Vlans allowed and active in management domain
Fa0/4     1,20,99

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/4     1,20,99

```

Screenshot 5: DNS-WebServer Static IP Configuration

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

FastEthernet0

Port Status
Link Speed
Duplex
MAC Address

100 Mbps
10 Mbps
Half Duplex
Full Duplex

☒ On
☒ Auto
☒ Auto

0060.4774.2DE4

IP Configuration
☒ DHCP
☒ Static

IPv4 Address
Subnet Mask

192.168.1.10
255.255.255.240

IPv6 Configuration
☒ Automatic
☒ Static

IPv6 Address
Link Local Address:

FE80::260:47FF:FE74:2DE4

NETWORK WORKFLOW & DATA FLOW

Expected Packet Flow (HR-PC1 → DNS-WebServer):

1. HR-PC1 initiates ping to 192.168.1.10
 - HR-PC1 IP: 192.168.3.5 (DHCP from R4)
 - Gateway: 192.168.3.1 (R4 Gig0/0.20)
 - Sends ARP: Who has 192.168.3.1?
2. Switch learns and forwards
 - SW-Floor2 Fa0/1 receives frame (untagged, in VLAN 20)
 - SW-Floor2 forwards to Fa0/4 (trunk port)
 - Frame tagged with VLAN 20 (802.1q header)
3. Router receives on trunk
 - R4 Gig0/0 receives tagged frame
 - Subinterface Gig0/0.20 de-tags frame
 - IP routing decision: 192.168.1.0/28 via 192.168.0.9 (OSPF)
4. Router forwards to R2
 - R4 Serial0/3/0 sends to R2 Serial0/2/0 (backbone)
 - R2 receives packet
 - R2 routing decision: 192.168.1.0/28 directly connected via Gig0/0.99
5. R2 forwards to server VLAN
 - R2 Gig0/0.99 forwards to SW-Servers (trunk)
 - Frame tagged with VLAN 99
 - SW-Servers Fa0/4 receives tagged frame
 - SW-Servers Fa0/1 de-tags and forwards to DNS-WebServer
6. Return path (reverse)
 - DNS-WebServer replies with source 192.168.1.10
 - Retraces path back through switches and routers

Why It's Failing:

One of these steps is breaking. Most likely candidates:

- Step 2: SW-Floor2 not tagging frames properly on trunk
- Step 3: R4 Gig0/0.20 not receiving de-tagged frames
- Step 6: Return traffic not reaching HR-PC1 (asymmetric)

COMPLETE SYSTEM ARCHITECTURE

Network Topology Components:

- 4 Cisco 2911 Routers (R1-Edge, R2-Core, R3-Floor1, R4-Floor2-3)
- 4 Cisco 2960-24TT Switches (SW-Floor1, SW-Floor2, SW-Floor3, SW-Servers)
- 6 End-user PCs (2 per floor)
- 1 DNS/Web Server (static IP in VLAN 99)
- 1 Access Point (for potential wireless extension)

Backend Infrastructure (Not shown in Packet Tracer):

- Flask REST API (Python) - Running on host machine
- SQLite Database - Stores ingested network anomalies
- CICIDS Dataset - Labeled network attack traffic
- ML Model (TBD by P3) - Trains on CICIDS data
- Web Dashboard (TBD by P4) - Visualizes detected anomalies

Future Extension (Netmiko Automation):

- Cisco DevNet Sandbox (IOS XE on Cat8kv)
- Netmiko Python library - Automate ACL/route configuration
- Closes detection-to-response loop: ML detects anomaly → Netmiko blocks traffic

RECOMMENDED NEXT STEPS

1. Immediate Debugging (Priority 1):

- Run 'ipconfig' on HR-PC1 → Verify gateway is 192.168.3.1
- Run 'tracert 192.168.1.10' from HR-PC1 → Show which hop fails
- Enable 'debug ip packet detail' on R4 → Capture packets on Gig0/0.20
- Check switch port counters on SW-Floor2 Fa0/4 → Verify traffic arriving
- Verify spanning-tree status on trunks → Ensure not blocked

2. If Still Failing (Priority 2):

- Check if Gig0/0.1 exists on any router (conflicting subinterface)
- Verify trunk allowed VLANs on switches include all required VLANs
- Test with static IP instead of DHCP to rule out gateway issues
- Check if native VLAN 99 is properly configured on both trunk sides
- Use packet capture (if available) to see what's happening on wire

3. Once Inter-VLAN Works (Priority 3):

- Test all PC-to-PC communication across VLANs
- Verify DNS-WebServer responds to ping from all VLANs
- Document final working configuration
- Move to Netmiko automation on DevNet Sandbox
- Create automated ACL deployment based on ML anomaly detection

4. Project Completion:

- P3 trains ML model on CICIDS data from database
- P4 builds dashboard to visualize anomalies
- Integration testing: Network captures → API → ML → Dashboard
- Final project report and presentation

CISCO INTERVIEW PREPARATION NOTES

Key Learning Points from This Project (For Interview):

1. VLAN & Inter-VLAN Routing:

Q: How do you enable communication between different VLANs?

A: "You need a Layer 3 device (router) configured with subinterfaces, one per VLAN, using dot1Q encapsulation. Each subinterface gets its VLAN's gateway IP. The switch trunks carry traffic tagged with VLAN IDs; the router de-tags on subinterfaces and performs IP routing between them. This is called 'router-on-a-stick.'"

2. DHCP in Multi-VLAN Environment:

Q: How does DHCP work across VLANs?

A: "DHCP requests are VLAN-local by default. You configure DHCP pools on routers for each VLAN, with the router's subinterface IP as the gateway. The router becomes the DHCP server for each VLAN. If you want a central DHCP server, you use DHCP relay (ip helper-address) to forward requests between VLANs."

3. Trunk Port Configuration:

Q: What's the difference between access and trunk ports?

A: "Access ports (switchport mode access) belong to one VLAN and send untagged frames. Trunk ports (switchport mode trunk) carry multiple VLANs, tagging each frame with 802.1q header. Native VLAN is sent untagged on trunk for backward compatibility."

4. Subinterface Encapsulation:

Q: Why use encapsulation dot1Q on subinterfaces?

A: "Encapsulation dot1Q tells the router how to interpret VLAN tags arriving on the trunk. Each subinterface is mapped to one VLAN ID. The router de-tags incoming frames on the matching subinterface and processes IP routing."

5. Debugging Inter-VLAN Issues:

Q: Your inter-VLAN routing isn't working. How do you troubleshoot?

A: "I follow this systematic approach: (1) Verify VLAN configuration on switches — show vlan brief. (2) Verify trunk ports are UP — show interfaces trunk. (3) Check router subinterfaces are UP — show ip interface brief. (4) Verify OSPF/routing — show ip route. (5) Use debug ip packet detail on router to see if packets arrive. (6) Check switch port counters to see if traffic hits the trunk. (7) Verify gateway IPs match what DHCP assigns."

Your Trained Answer (For This Project):

"I designed and manually configured a 4-router, 4-switch network topology with four VLANs using Cisco Packet Tracer. I implemented OSPF routing across the backbone and configured router-on-a-stick inter-VLAN routing using subinterfaces with dot1Q encapsulation. Each VLAN has DHCP on the gateway router. Currently debugging inter-VLAN routing failure where cross-VLAN traffic drops silently — likely a trunk port configuration issue. My systematic approach: verify VLAN/trunk/subinterface configs, check routing tables, use debug ip packet to identify where packets drop. Once working, I'm extending this with Netmiko on DevNet Sandbox to automate ACL deployment when ML model detects network anomalies — closing the detection-to-response loop which is exactly how modern SOC's work."

CONCLUSION

Current Status Summary:

The Network Anomaly Detection System project is 60% complete. Phases 1-2 (IP addressing and OSPF routing) are fully working, and Phase 3 (VLANs) is partially complete — DHCP is operational for all VLANs, but inter-VLAN traffic is currently failing. This is a common networking challenge that requires systematic packet-level debugging to identify the exact failure point.

What Makes This Project Valuable for Interview:

1. End-to-end network design from scratch (topology, IP planning, routing protocols)
2. Hands-on Cisco CLI configuration (2911, 2960)
3. Problem-solving under incomplete information (debugging without packet capture)
4. Integration of network with ML/backend systems (not just network in isolation)
5. Demonstrates knowledge of modern DevOps practices (Netmiko automation planned)

Project Resilience:

Even with the current routing failure, the project demonstrates solid networking fundamentals. The ability to quickly identify that "DHCP works, OSPF works, interfaces are UP, but traffic doesn't flow" is exactly the kind of diagnostic thinking Cisco interviewers respect. The next phase (DevNet Sandbox with Netmiko) will add significant value by showing how to automate response to network events — a critical skill in modern network engineering.

Completion Timeline:

- Phase 3 debugging: 1-2 hours (once root cause identified)
- Phase 4 Netmiko automation: 2-3 days
- Final integration & testing: 1 week
- Project completion: Week 4 (On schedule)

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Next Review: After debugging steps are executed